Extending thinking about sequences

Task A: Concrete versus abstract sequences

1) This table shows us the history of men's 100 m sprint world record times.

Year	1940	1960	1980	2000	2020	2040	2060	2080	2100
Time (s)	10.22	10.06	9.9	9.74	9.58	9.42	9.26	9.1	8.94

-0.16 -0.16 -0.16 -0.16 -0.16 -0.16 -0.16 -0.16

a) Identify a pattern and use it to predict the times up to the year 2100.

Arithmetic sequence of common difference -0.16

b) This is a concrete sequence. What is its physical limit?

The year 3140 *would see a time of* -0.1 *seconds. It would be impossible to have a negative term in this context. The sequence would become abstract.*

Task B: Extending sequences in both directions

1) Scientists start studying the Penguin population on an island in the Southern Ocean annually and they find these numbers.

Year	1	2	3	4	5	6
Population (000's)	160	200	250	312.5	390.6	488.3

$\times 1.25$ $\times 1.25$ $\times 1.25$ $\times 1.25$ $\times 1.25$

$200 \div 160 = 1.25$

The population data forms a geometric sequence.

a) Move forwards to estimate the population next year and in two years time.

$312.5 \times 1.25 = 390.625$

$390.625 \times 1.25 = 488.281\ldots$

b) Move backwards to find how many years ago the population was under one hundred thousand.

Assume this model worked historically.

$160 \div 1.25 = 128$

$128 \div 1.25 = 102.4$

$102.4 \div 1.25 = 81.92$

The population was under one hundred thousand three years before the study began.